Taehyun Hwang

Last updated: August 2026

Contact Information

Graduate School of Data Science
Seoul National University
1 Gwanak-ro, Gwanak-gu, Seoul, South Korea, 08826

th.hwang@snu.ac.kr
taehyunable.hwang@gmail.com
https://taehyunable.com

Positions

- Seoul National University — Mar. 2025 – present
  Postdoctoral Researcher, Research Institute for Data Science / Graduate School of Data Science
- Sungkyunkwan University — Sep. 2018 – Feb. 2020
  Research Assistant, Institute of Basic Science

Education

- **Seoul National University** — Mar. 2020 – Feb. 2025
  Ph.D in Data Science (Advisor: Min-hwan Oh)
  Thesis: *Efficient Exploration for Sequential Decision Making with Non-linear Function Approximation*
- **Sungkyunkwan University** — Sept. 2016 – Aug. 2018
  M.S in Mathematics (Advisor: Yongdo Lim / Co-advisor: Rhee Man Kil)
  Thesis: *Combination of Gaussian Kernel Function Networks for Traffic Flow Prediction*
- **Kyungpook National University** — Mar. 2010 – Aug. 2016
  B.S. in Mathematics, with a minor in Business Administration (Graduated with the highest honor)

Research Interests

My research lies at the intersection of machine learning, sequential decision-making, and applied mathematics, with a particular focus on **reinforcement learning**, **contextual bandits**, **online learning**, and **statistical machine learning**. I aim to develop *mathematically grounded* and *practically implementable* learning algorithms, and to understand the fundamental limits of learning and optimization under uncertainty. Ultimately, I am interested in applying these ideas to a broad range of real-world decision-making problems.

Publications (* denotes equal contribution)

9. **Generalized Linear Bandits with Memory.**
   Heesang Ann*, Hyunjun Choi*, **Taehyun Hwang***, Younghoon Shin, Haeju Cheong, and Min-hwan Oh
   *International Conference on Machine Learning (**ICML**)*, 2026

8. **Diversified Multinomial Logit Contextual Bandits.**
   Heesang Ann, **Taehyun Hwang**, and Min-hwan Oh
   *The Fourteenth International Conference on Learning Representations (**ICLR**)*, 2026
   - Outstanding Paper Award, Korean Artificial Intelligence Association 2025 Fall Conference

7. **Tractable Multinomial Logit Contextual Bandits with Non-Linear Utilities.**
   Taehyun Hwang, Dahngoon Kim, and Min-hwan Oh
   *Advances in Neural Information Processing Systems 39 (**NeurIPS**)*, 2025
   - Best Paper Award, Korean Artificial Intelligence Association 2025 Summer Conference

6. **Lasso Bandit with Compatibility Condition on Optimal Arm.**
   Harin Lee*, **Taehyun Hwang***, and Min-hwan Oh
   *The Thirteenth International Conference on Learning Representations (**ICLR**)*, 2025
   - Outstanding Paper Award, Korean Artificial Intelligence Association 2024 Summer Conference
   - NRF President's Award (2nd Prize), 2024 K-DS Conference Research Presentation

5. **Randomized Exploration for Reinforcement Learning with Multinomial Logistic Function Approximation.**
   Wooseong Cho*, **Taehyun Hwang***, Joongkyu Lee, and Min-hwan Oh
   *Advances in Neural Information Processing Systems 38 (**NeurIPS**)*, 2024
   • Outstanding Paper Award, Korean Artificial Intelligence Association 2023 Summer Conference

4. **Model-based Reinforcement Learning with Multinomial Logistic Function Approximation.**
   Taehyun Hwang and Min-hwan Oh
   *Proceedings of the 37th AAAI Conference on Artificial Intelligence (**AAAI**), 2023.*
   • Oral presentation in the main technical session

3. **Combinatorial Neural Bandits.**
   Taehyun Hwang*, Kyuwook Chai*, and Min-hwan Oh
   *Proceedings of the 40th International Conference on Machine Learning (**ICML**), 2023*
   • Outstanding Paper Award, Korean Artificial Intelligence Association 2022 Summer Conference

2. **Deep Learning Aided Evaluation for Electromechanical Properties of Complexly Structured Polymer Nanocomposites.**
   Kyungmin Baek, **Taehyun Hwang**, Wonseok Lee, Hayoung Chung, and Maenghyo Cho
   Composites Science and Technology, 109661, 2022.

1. **Estimation of Jamming Parameters based on Gaussian Kernel Function Networks.**
   Taehyun Hwang, Rhee Man Kil, Hyun Ku Lee, Jung Ho Kim, Jae Heon Ko, Jeil Jo, and Junghoon Lee
   Journal of the Korea Institute of Military Science and Technology 23 (1), 1-10, 2020.

Preprints

4. **Linear Ensemble Sampling with Fewer Ensembles.**
   Taehyun Hwang and Min-hwan Oh
   Submitted

3. **Block Optimism for Nonstationary Bandits with Latent Linear Dynamics.**
   Taehyun Hwang, Hyunjun Choi, Heesang Ann, and Min-hwan Oh
   Submitted
   • Best Paper Award, Korean Artificial Intelligence Association 2026 Summer Conference

2. **Topic-Aware Contextual Cascading Bandits.**
   Hyunjun Choi*, **Taehyun Hwang***, and Min-hwan Oh
   Submitted

1. **Sample-Efficient Pruning Model Selection via Lasso.**
   Kyuwook Chai*, **Taehyun Hwang***, and Min-hwan Oh
   Submitted

Grants

- **Flexible Deep Neural Network-based Bandit Algorithm for General-purpose Use.**
  Ascending SNU Future Leader Fellowship May. 2025 – Apr. 2026

Project Experience

- **Development of Analysis Model to Explore Test Process Equipment Combination and Improve Flexible Test Performance.**
  Director: Prof. Min-hwan Oh Mar. – Sep. 2022
  Funded by *SK hynix*

- **A Study on the Algorithms for Estimating the Parameters of Autonomous Jamming Technique.**
  Director: Prof. Rhee Man Kil Apr. – Dec. 2018
  Funded by *Agency for Defense Development* and *LIG Nex1*

- **An Industrial Problem Solving Project Using Mathematical Data Analysis.**
  Director: Prof. Yongdo Lim — May – Sep. 2017
  Funded by *National Institute for Mathematical Sciences*

Awards & Honors

- **Best Paper Award**, Korean Artificial Intelligence Association — Aug. 2026
- **Gold Reviewer**, ICML — May 2026
- **Top Reviewer**, NeurIPS — Dec. 2025
- **Outstanding Paper Award**, Korean Artificial Intelligence Association — Nov. 2025
- **Best Paper Award**, Korean Artificial Intelligence Association — Aug. 2025
- **Top Reviewer**, ICML — Jul. 2025
- **Best Reviewer**, AISTATS — May 2025
- **NRF President's Award (2nd Prize)**, K-DS Conference Research Presentation — Nov. 2024
- **Outstanding Paper Award**, Korean Artificial Intelligence Association — Aug. 2024
- **YOULCHON AI Star Scholarship**, Youlchon Foundation — Aug. 2023
- **Outstanding Paper Award**, Korean Artificial Intelligence Association — July 2023
- **AAAI Student Scholarship**, AAAI — Feb. 2023
- **Outstanding Paper Award**, Korean Artificial Intelligence Association — Aug. 2022

Teaching Experience

- **Teaching Assistant** at Seoul National University
  Data Science & Reinforcement Learning, Foundation of Data Science, Math for Data Science
- **Teaching Assistant** at Sungkyunkwan University
  Calculus I/II, *Linear Algebra, Discrete Mathematics, Topics in Algebra (Lie Algebra)*

Invited Talks & Conference Presentations

"Contextual Combinatorial Bandits with Nonlinear Function Approximation"
- Korean Institute for Advanced Study (KIAS), Seoul — Mar. 2026

"Tractable Multinomial Logit Contextual Bandits with Non-Linear Utilities"
- K-DATA SCIENCE Conference Research Presentation, Daegu — Sep. 2025

"Efficient Exploration Algorithms for Reinforcement Learning with Multinomial Logistic Function Approximation"
- Kyungpook National University, Department of Mathematics — Nov. 2024

"*Sequential Decision Making Algorithms with Function Approximation*"
- INFORMS APS PhD Student Showcase, Seattle — Oct. 2024

"*Lasso Bandit with Compatibility Condition on Optimal Arm*"
- Korean AI Association (KAIA) Summer Conference, Pyeongchang — Aug. 2025
- Korea Computer Congress (KCC), Jeju — Jul. 2025
- INFORMS 2024 Annual Meeting, Seattle — Oct. 2024
- SNU-KAIST AL/ML Theory Workshop, Gangneung — Aug. 2024

"*Introduction to Online Decision Making*"
- Sungkyunkwan University, Department of Mathematics — Nov. 2023

"*Neural Contextual Bandits from Single Action to Combinatorial Actions*"
- SK Telecom Market Top AI Course, SK Telecom Tower — Jul. 2023

"*Model-based Reinforcement Learning with Multinomial Logistic Function Approximation*"
- Korea Software Congress (KSC), Busan — Dec. 2023
- Korea Data Mining Society (KDMS), Gangneung-Wonju National University — Jun. 2023
- AAAI 2023, Washington, D.C. — Feb. 2023
- KAIA & NAVER Joint Conference, NAVER 1784, Seongnam — Nov. 2022
- INFORMS 2022 Annual Meeting, Indianapolis — Oct. 2022

"*Combinatorial Neural Bandits*"
- Advanced Neuroimaging and AI 2024, Seoul National University — Jun. 2024
- AI Seoul 2024, Seoul City Hall — Feb. 2024
- Korean Statistical Society (KSS), Sungshin Women's University — Dec. 2023
- AIIS Retreat 2023, Seoul National University — Nov. 2023
- Korea Data Mining Society (KDMS), Busan — Aug. 2022
- Korean Artificial Intelligence Association (KAIA), Jeju — Aug. 2022
- INFORMS 2021 Annual Meeting, Anaheim (Virtual) — Oct. 2021

"*A Gaussian Kernel Function Network for Traffic Flow Prediction*"
- Korean Mathematical Society (KMS) 2019 Annual Meeting, Hongik University — Oct. 2019

"*Mathematics and Machine Learning*"
- Kongju National University, Department of Applied Mathematics — Sep. 2019

Academic Service

- **Reviewer and Program Committee Member**
  ICML (2023–2026)
  NeurIPS (2022–2026)
  ICLR (2024–2026)
  AAAI (2025–2027)
  AISTATS (2025–2026)